

Soln: Note that there are no perfect squares n with $10 \leq n \leq 15$, because $3^2 = 9$ and $4^2 = 16$. Hence the statement that n is an integer with $10 \leq n \leq 15$ which is a perfect square is false for all integers n . Consequently, the statement to be proved is true for all integers n . $\square$

We can also quickly prove a conditional statement $p \rightarrow q$ if we know that the conclusion q is true. By showing that q is true, it follows that $p \rightarrow q$ must also be true. A proof of $p \rightarrow q$ that uses the fact that q is true is called a **trivial proof**.

Example 7: Let $P(n)$ be "If a and b are positive integers with $a \geq b$, then $a^n \geq b^n$," where the domain consists of all nonnegative integers. Show that $P(0)$ is true.

Soln: The proposition $P(0)$ is "If $a \geq b$, then $a^0 \geq b^0$." Because $a^0 = b^0 = 1$, the conclusion of the conditional statement "If $a \geq b$, then $a^0 \geq b^0$ " is true. Hence, this conditional statement, which is $P(0)$, is true. This is an example of a trivial proof. Note that the hypothesis, which is the statement " $a \geq b$," was not needed in this proof. $\square$

Definition 2: The real number r is **rational** if there exists integers p and q with $q \neq 0$ such that $r = p/q$. A real number that is not rational is called **irrational**.

Example 8: Prove that the sum of any two arbitrary rational numbers is rational.

Solution: Let r and s be two arbitrary rational numbers. By Defn 2, there exists integers p_1, q_1, p_2, q_2 , where $q_1 \neq 0$ and $q_2 \neq 0$ such that $r = \frac{p_1}{q_1}$ and $s = \frac{p_2}{q_2}$. $\therefore r + s = \frac{p_1}{q_1} + \frac{p_2}{q_2} = \frac{p_1 q_2 + p_2 q_1}{q_1 q_2}$